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JeongJin Kim¹, Jiyoung Park², Young Woo Sohn¹, and Jung In Lim¹

Abstract

Despite increasing research on the mechanisms that underlie the relation between perceived overqualification (POQ) and extra-role behaviors, the relation's affective mechanisms have been understudied. In this study, we focus on the role of job boredom among overqualified workers in order to understand two types of extra-role behaviors: counterproductive work behaviors (CWB) and organizational citizenship behaviors (OCB). Using a sample of white-collar employees in South Korea ($N = 298$), we found that job boredom fully mediated the effects of POQ on CWB and OCB after controlling for the effects of state anger and neuroticism. We additionally examined whether career calling moderates the indirect effects of POQ on CWB and OCB via job boredom and found career calling mitigated the negative indirect effect of POQ on OCB via job boredom. However, career calling did not moderate the positive indirect effect of POQ on CWB. Implications of these findings are discussed.

Keywords

perceived overqualification, counterproductive work behavior, organizational citizenship behavior, job boredom, career calling

Organizational scholars and practitioners have devoted considerable attention to studying the job attitudes and behaviors of overqualified workers (Erdogan & Bauer, 2009; Liu, Luksyte, Zhou, Shi, & Wang, 2015), as perceptions of overqualification have become a globally prevalent phenomenon (Erdogan & Bauer, 2009; McKee-Ryan & Harvey, 2011). Defined as the degree to which workers perceive themselves as possessing a surplus of skills, education, and experience compared to their job requirements (Maynard, Joseph, & Maynard, 2006), perceived overqualification (POQ) can lead to unfavorable outcomes for both employees and employers (Harari, Manapragada, & Viswesvaran, 2017). Workers who view themselves as overqualified tend to suffer from lower psychological well-being (Johnson & Johnson, 1996), show less commitment to the organization (Feldman, Leana,

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& Bolino, 2002), and engage in more withdrawal behaviors (Erdogan & Bauer, 2009; Maynard & Parfyonova, 2013). Extra-role behaviors, active and voluntary behaviors beyond the core tasks (Miles, Borman, Spector, & Fox, 2002), were also predicted by employees' level of POQ. Overqualified workers tend to be negligent in extra-role behavior such that they tend to demonstrate more counterproductive work behaviors (CWB; Liu et al., 2015; Luksyte, Spitzmueller, & Maynard, 2011) and fewer organizational citizenship behaviors (OCB; Deng et al., 2018).

Integrating person-job (P-J) fit theory (Kristof-Brown & Guay, 2011) and emotion-centered model of voluntary work behavior (Spector & Fox, 2002), we focus on the role of boredom at work in explaining why POQ is related to CWB and OCB. Boredom occurs in response to understimulation at work, which overqualified workers often perceive (Spector & Fox, 2010). To investigate the underlying mechanisms of overqualified employees' CWB and OCB, prior studies have found that several cognitive factors, such as task significance, person-group fit (Hu et al., 2015), P-J fit, job burnout (Luksyte et al., 2011), organization-based self-esteem (Liu et al., 2015), and social acceptance (Deng et al., 2018), account for the effects of POQ on extra-role behaviors.

However, the specific affect type that underlies this process remains understudied. Organization is an environment that induces strong emotions, and the individual's specific emotion induces certain action tendencies and leads to behaviors (Spector & Fox, 2002). Boredom has received increasing attention in understanding job behaviors as employees frequently experience job boredom in diverse industries (Fisher, 1993; Harju, Hakanen, & Schaufeli, 2014). Boredom triggers impulses to escape the situation and elicit coping behaviors that reduce this boredom (Spector & Fox, 2002). As a coping mechanism to deal with boredom, bored employees tend to perform behaviors in a counterproductive manner (Spector & Fox, 2010) and underperform in extra-role behaviors (i.e., CWB and OCB; Bruursema, Kessler, & Spector, 2011; Fisher, 1993). Boredom can play a key role in understanding workers' CWB and OCB (Liu & Wang, 2012; Spector & Fox, 2010), but its mediating role of linking POQ with CWB and OCB has not been examined empirically.

Moreover, we investigate an individual factor that influences the relations between job boredom and the two types of behaviors. We suggest that career calling, which refers to a transcendent summons that is oriented toward demonstrating prosocial values and meaningfulness (Dik & Duffy, 2009), can buffer the negative effects of job boredom. Due to prosocial intention of career calling, it motivates people to persist in performing work beyond their duty and engaging in prosocial and moral behaviors even when they experience negative emotions (Bunderson & Thompson, 2009; Schabram & Maitlis, 2017).

Based on P-J fit theory and emotion-centered model of CWB and OCB, we aim to examine the associations of job boredom with CWB and OCB and the mediating role of job boredom on the relations between POQ and the two extra-role behaviors. To our knowledge, our study offers the first empirical evidence of boredom at work as the affective mechanism of the focal relations. We also test a moderated mediation model in which career calling moderates the indirect effect of POQ on CWB and OCB via job boredom. With the findings, we expect to offer insights on when and why POQ relates to CWB and OCB.

POQ and Job Boredom: P-J Misfit

Overqualification refers to the degree to which employees possess surplus knowledge, skills, and abilities (KSAs) relative to their job requirements (Erdogan, Bauer, Peiró, & Truxillo, 2011; Maynard et al., 2006). Objective overqualification captures whether an employee's qualifications exceed job demands and is often judged by someone else (e.g., a coworker or supervisor) rather than by the employee, while subjective overqualification refers to the degree to which employees consider themselves overqualified and is assessed using self-report measures (Maltarich, Reilly, & Nyberg, 2011). Most studies employ a subjective measure, since POQ is deemed to be a more proximal indicator

of job attitudes and behaviors (Erdogan et al., 2011; Liu & Wang, 2012). Erdogan, Bauer, Peiró, and Truxillo (2011) argue that subjective overqualification is more suitable for psychological research because objective measures of whether one is overqualified are “by necessity deficient for psychological research” (p. 218) and tend to focus on a limited range of abilities, skills, and credentials required by the job. Thus, following prior studies, we employ a perceptual measure of overqualification to examine the underlying process by which workers feel overqualified for their jobs.

Building on P-J fit theory and prior work, we believe that POQ can be understood in the lens of P-J fit (e.g., Liu et al., 2015). P-J fit refers to the match or congruence between the person and the job (Kristof-Brown & Guay, 2011). POQ is the perception of the incompatibility between their KSAs (i.e., person) and job demands (i.e., job). According to the P-J fit literature, poor P-J fit is associated with lower job satisfaction, greater intention to quit, and poorer overall performance, which includes extra-role performance (Kristof-Brown, Zimmerman, & Johnson, 2005). Similarly, overqualified employees tend to show poor job attitudes and extra-role performance in terms of CWB and OCB (Feldman, 1996). Namely, they are likely to show more CWB (Liu et al., 2015; Luksyte et al., 2011) and more OCB (Maynard & Feldman, 2011).

P-J fit theories state that when the employee perceives P-J misfit, negative affect is experienced such as job boredom (Edwards & Van Harrison, 1993). Job boredom is defined as an unpleasant and deactivated affect characterized by low arousal and dissatisfaction stemming from an understimulating work environment (Fisher, 1993; Reijseger et al., 2013). Given that job boredom is an emotional reaction to an understimulating work environment, it could be commonly experienced by workers who feel that their abilities are underutilized or those who perceive a mismatch between their skills and job requirements (Liu & Wang, 2012). Consistent with this argument, using a sample of health-care workers, Watt and Hargis (2010) found a positive association between POQ and boredom proneness. Boredom is also positively associated with work underload and skill underutilization (Caplan, Cobb, French, van Harrison, & Pinneau, 1975). Based on these findings, we hypothesized a positive relation between POQ and job boredom.

Hypothesis 1: POQ is positively related to job boredom.

CWB and OCB as Consequences of Job Boredom

Our study builds on the emotion-centered model of voluntary work behaviors (Spector & Fox, 2002) by showing job boredom as an affective path from personal appraisal (e.g., P-J fit) to CWB and OCB. CWB is defined as voluntary behaviors that damage the well-being of the organization and other individuals (Bennett & Robinson, 2000). According to the emotion-centered model of voluntary work behaviors (Spector & Fox, 2002), negative emotions caused by workplace stressors can trigger CWB. In support of this model, Bruursema, Kessler, and Spector (2011) found job boredom was positively related to different types of CWB including abuse, sabotage, and withdrawal. The positive relation between boredom and CWB has been supported across diverse occupations (Drory, 1982; Spector et al., 2006; van Hooff & van Hooft, 2014). Based on these findings, we expected that workers who feel bored would be likely to engage in CWB.

Hypothesis 2: Job boredom is positively related to CWB.

OCB refers to discretionary behaviors beneficial to the organization and its members, such as helping others or attending events that are not mandatory (Lee & Allen, 2002). The emotion-centered model of CWB and OCB (Spector & Fox, 2002) connects OCB only to positive affect, but our study builds on the model by demonstrating the relation between negative affect (i.e., job boredom) and OCB. Bored individuals may not be willing to help others, since boredom is linked with low sociability

(Leong & Schneller, 1993) and alienation (Tolor, 1989), indicating less interpersonal OCB. Job boredom can result in complacency regarding tasks and social interactions, which may dampen responses to novel situations (Liu & Wang, 2012). Those with high sensitivity to social stimuli tend to show more prosocial behaviors (Chiaburu, Oh, Berry, Li, & Gardner, 2011), but bored employees experience low alertness and tend to engage in withdrawal and nonfunctional behaviors (van Hooft & van Hooft, 2014), thereby less likelihood of behaviors helpful to the organization.

Other researchers argue that bored workers would engage in OCB in order to escape from boredom (Liu & Wang, 2012; Spector & Fox, 2010) because boredom drives novelty-seeking behaviors (Schweizer, 2006). However, bored individuals tend to escape the situation by performing sensation-seeking and high-arousing behaviors that are potentially harmful deeds to self and others (Elpidorou, 2017). Given that there are multiple alternatives other than OCB at work, we believe that boredom would relate negatively to OCB. Additionally, as a host of studies have shown that job satisfaction is a robust predictor of OCB (Organ & Ryan, 1995; Whitman, Van Rooy, & Viswesvaran, 2010), feeling bored at work—the opposite of job satisfaction (Reijseger et al., 2013)—is unlikely to motivate workers to perform OCB. Bored workers exhibit negative attitudes toward the organization, such as greater intention to leave (Harju et al., 2014) and lower work engagement (Harju, Hakanen, & Schaufeli, 2016), all of which are associated with less engagement in OCB (Saks, 2006). Thus, we hypothesized that bored individuals are less likely to perform OCB.

Hypothesis 3: Job boredom is negatively related to OCB.

The Affective Process of Overqualified Workers: The Role of Job Boredom

Negative emotions serve as a critical factor in understanding the CWB and OCB of overqualified individuals (Liu & Wang, 2012). Using time-lagged data on Chinese workers, Liu et al. (2015) found that overqualified employees experienced anger toward their employment situation and that their level of anger predicted both self-rated and supervisor-rated CWB. Another negative emotional reaction, cynicism about their job, has also been found to be a reason overqualified individuals engage in CWB (Luksyte et al., 2011). Specifically, when including multiple potential mediators, such as needs-supplies fit and psychological contract, only cynicism fully mediated the relation between POQ and CWB, which highlights the importance of negative affect in understanding the focal relation (Luksyte et al., 2011). With regard to the effect of negative emotions on OCB, Deng et al. (2018) conducted two studies using multisource data and found that when overqualified employees were less interpersonally skilled, they felt less socially accepted, and this unpleasant feeling reduced their OCB performance. In experimental studies, Belschak and Den Hartog (2009) reported that positive emotions increased participants' intention to engage in OCB, while negative emotions decreased their intention to engage in OCB. Based on these findings, we expect that POQ is positively related to CWB but negatively related to OCB.

Among negative emotions, boredom has recently received attention as a distinct emotional state that significantly influences work behaviors (van Hooft & van Hooft, 2014). Research suggests that boredom can mediate the relation of POQ with CWB and OCB. Liu and Wang (2012) argue that boredom can be a frequent emotional reaction to POQ and ultimately lead to maladaptive work behaviors. Similarly, Spector and Fox (2010) suggest that feeling bored at work due to an understimulating work environment can influence CWB and OCB. Erdogan, Bauer, Peiró, and Truxillo (2011) assert that because of the perceived misfit between their KSAs and jobs, overqualified workers who feel a lack of stimulation in the workplace may experience boredom at work, which, in turn, causes them to engage more in CWB and less in OCB. Research on POQ stresses that boredom may play a key role in explaining CWB and OCB of overqualified people (Liu & Wang, 2012; Spector & Fox, 2010).

Taken together, we hypothesized that boredom would mediate the associations between POQ and CWB and OCB.

Hypothesis 4a: Job boredom mediates the positive relation between POQ and CWB.

Hypothesis 4b: Job boredom mediates the negative relation between POQ and OCB.

The Moderating Role of Career Calling

Researchers have highlighted the need for studies examining the potential moderators that may impact the relations between job boredom and CWB and OCB (Bruursema et al., 2011). Although boredom triggers certain emotional tendencies, its effects on behaviors vary depending on individual and contextual factors (Spector & Fox, 2002). The degree to which they perceive their work as a career calling can influence how people behave in response to negative emotions (Schabram & Maitlis, 2017; Ugwu & Onyishi, 2018). One of the salient features of career calling is prosocial intention to contribute to greater good through work (Elangovan, Pinder, & McLean, 2010). As do others without a career calling, those with a career calling also feel negative emotions toward challenging tasks and situations, some of which are intense due to strong identification with work (Schabram & Maitlis, 2017). Despite the unpleasant feelings, people with callings are more likely to behave in constructive and other-focused manners focusing on the purpose and meaning of their work (Bunderson & Thompson, 2009; Schabram & Maitlis, 2017). A moderating role of career calling on the relation between negative emotion, such as frustration, and work engagement was found (Ugwu & Onyishi, 2018), and an empirical study has supported the positive link between career calling and OCB among Korean salespersons (Park, Sohn, & Ha, 2016). Despite the absence of empirical research demonstrating the relation between career calling and CWB, the fact that the prosocial intention of career calling drives people to perform beyond their duty and to engage in more moral and prosocial behaviors at work (Bunderson & Thompson, 2009) suggests a negative relation between calling and CWB. Based on these findings, we expected that career calling would mitigate the negative effects of job boredom on CWB and OCB.

Hypothesis 5a: Career calling moderates the effect of job boredom on CWB such that the positive relation between job boredom and CWB becomes weaker (vs. stronger) when employees perceive a higher (vs. lower) sense of career calling.

Hypothesis 5b: Career calling moderates the effect of job boredom on OCB such that the negative relation between job boredom and OCB becomes weaker (vs. stronger) when employees perceive a higher (vs. lower) sense of career calling.

Considering the buffering effect of career calling, we also tested for moderated mediation hypotheses to determine whether career calling would moderate the indirect effect of POQ on CWB and OCB via job boredom.

Hypothesis 6a: Career calling moderates the positive indirect effect of POQ on CWB via job boredom such that this indirect effect is weaker (vs. stronger) for employees who perceive a higher (vs. lower) sense of career calling.

Hypothesis 6b: Career calling moderates the negative indirect effect of POQ on OCB via job boredom such that this indirect effect is weaker (vs. stronger) for employees who perceive a higher (vs. lower) sense of career calling.

Method

Participants and Procedure

We recruited 300 South Korean white-collar employees ranging in age from 18 to 39 years via an online survey. Our sampling criteria included age restriction in order to garner a sample that is prone to feeling boredom, as previous research reports a negative relation between boredom and age (Drory, 1982; Harju et al., 2014). In addition, as boredom is known to be associated with repetitive and monotonous tasks (Fisher, 1993), early research in work-related boredom mainly studied blue-collar workers (e.g., Drory, 1982). However, more recent research has shown that the experience of job boredom can also be prevalent among white-collar workers (Fisher, 1993; Van der Heijden, Schepers, & Nijsen, 2012), and thus we recruited a sample of white-collar workers to examine their experience of job boredom and its consequences. Of the 300 participants, 2 were omitted due to careless responses, and there were no missing data. After completing the online survey, participants were paid approximately 1.5 U.S. dollars.

Participants consisted of 149 men and 149 women, with an average age of 34 ($M = 33.88$, $SD = 4.18$) and average job tenure of 101.40 months ($SD = 46.59$). Their educational levels included associate's degree (17.8%), bachelor's degree (66.8%), master's degree (7.7%), and doctoral degree (0.1%). Participants worked in a variety of industry sectors, including business, medical services, information technology, education and social welfare, and manufacture.

We controlled for the effects of neuroticism and anger, as previous research has consistently confirmed that neuroticism is related positively to CWB (e.g., Berry, Ones, & Sackett, 2007) and negatively to OCB (e.g., Chiaburu et al., 2011). Also, state anger has been found to mediate the relation between POQ and CWB (Liu et al., 2015). By controlling for neuroticism and state anger, we were able to tease out the effects of an emotional tendency to negative affect and a possible affective mechanism in our model from the true effect of POQ.

Measures

For measures of neuroticism, state anger, CWB, OCB, and career calling, we used existent Korean versions of scales that are validated by previous work with Korean samples. For measures of POQ and job boredom, we translated the measures using a back-translation method (Brislin, 1970).

POQ. POQ is assessed using 9 items from the Scale of Perceived Overqualification (Maynard et al., 2006), which has been validated and used in prior studies (e.g., Maynard et al., 2006; Maynard & Parfyonova, 2013). Using a 5-point Likert-type scale (1 = *strongly disagree* to 5 = *strongly agree*), participants rated the extent to which they perceived themselves as possessing a surplus of KSAs compared to their job requirements. A sample item is "My job requires less education than I have." In this study, the Cronbach's α was .86.

Job boredom. Job boredom was assessed by the 6-item Dutch Boredom Scale (Reijseger et al., 2013), which was validated and used in numerous studies (e.g., Harju et al., 2016). Using a 7-point Likert-type scale (0 = *strongly disagree* to 6 = *strongly agree*), participants were asked to rate the extent to which they had felt bored at their jobs during the past month. A sample item is "I have felt bored at my job." In this study, the Cronbach's α was .85.

CWB. CWB was measured using Bennett and Robinson's (2000) 19-item measure of workplace deviance, which has been validated and used in previous studies (e.g., Berry et al., 2007). On a 7-point Likert-type scale (1 = *strongly disagree* to 7 = *strongly agree*), participants were asked to rate

the extent to which they have engaged in CWB during the past month. A sample item is “Said something hurtful to someone at work.” In this study, the Cronbach’s α was .93.

OCB. OCB was assessed using Lee and Allen (2002)’s 16-item measure, which has been validated and used in numerous studies (e.g., Chiaburu et al., 2011). On a 7-point Likert-type scale (1 = *strongly disagree* to 7 = *strongly agree*), participants rated their OCB during the past month. A sample item is “Help others who have been absent.” In this study, the Cronbach’s α was .92.

Career calling. Career calling was measured using a 12-item Calling and Vocation Questionnaire (Dik, Eldridge, Steger, & Duffy, 2012), which has been validated and used in numerous studies (e.g., Park et al., 2016). On a 4-point Likert-type scale (1 = *not at all true of me* to 4 = *absolutely true of me*), participants were asked to rate how much they perceive a career calling in their work. A sample item is “I believe that I have been called to my current line of work.” In this study, the Cronbach’s α was .89.

Control variables. We controlled for the effects of neuroticism and state anger, along with demographic variables. Neuroticism was measured using 3 items from the Big Five Inventory (John & Srivastava, 1999), which has been validated and used in other studies (e.g., Whiteoak, 2014). On a 5-point Likert-type scale (1 = *strongly disagree* to 5 = *strongly agree*), a sample item is “I see myself as someone who worries a lot.” The Cronbach’s α was .82 in this study. State anger was assessed using 10 items from State-Trait Anger Expression Inventory (Spielberger, 1988), which has been validated and used in prior studies (e.g., Chu, 2014). On a 4-point Likert-type scale (0 = *almost never* to 4 = *very often*), participants were asked to respond to such an item as “I have felt furious” during the past month. In this study, the Cronbach’s α was .94.

Analytic Procedure

To ensure that psychometric properties and factor loadings were appropriate, we examined a latent variable measurement model using Mplus 6.0 and then tested our mediation hypotheses by estimating the structural model. We used bootstrapping analyses and obtained 95% bias-corrected bootstrap confidence intervals (CIs) from 5,000 samples. To examine the moderation hypotheses, we conducted hierarchical multiple regression analyses using IBM SPSS Statistics Version 22. Lastly, we used the PROCESS macro for SPSS (Model 14; Hayes, 2013) to test the moderated mediation hypotheses.

Results

To assess the construct validity of our main constructs, we tested the measurement models by conducting a set of CFAs (confirmatory factor analyses). We estimated a six-factor model (i.e., POQ, anger, neuroticism, job boredom, CWB, and OCB), which showed a good fit with the data, $\chi^2(120) = 309.04$, $p < .001$, CFI = .95, SRMR = .05, RMSEA = .07, and all of the factor loadings were significant (all $ps < .001$). Thus, we proceeded to test the structural model using the six-factor model.

Table 1 presents descriptive and correlational findings. Consistent with our predictions, we found a positive correlation between POQ and boredom. Job boredom was positively correlated with CWB but negatively correlated with OCB. Career calling was negatively correlated with job boredom but positively with OCB. Next, the hypothesized mediation model yielded an acceptable fit to the data, $\chi^2(158) = 361.36$, $p < .001$, CFI = .95, SRMR = .04, RMSEA = .07. Overall, all of the hypothesized relations in the mediation model were significant. As expected, POQ was positively related to job boredom ($\beta = .29$, $p < .001$), supporting Hypothesis 1. Hypotheses 2 and 3 were supported as job boredom was positively related to CWB ($\beta = .26$, $p < .001$) and negatively related to OCB ($\beta = -.21$, $p < .05$). As for the mediation hypotheses, our results revealed that job boredom fully mediated the relations

Table 1. Descriptive Statistics, Correlations, and Reliabilities (Cronbach's α s in parentheses) among Study Variables.

Variables	<i>M</i>	<i>SD</i>	1	2	3	4	5	6	7	8	9	10
1. Age	33.88	4.18	—									
2. Gender	0.50	0.50	-.35**	—								
3. Job tenure	101.40	46.59	.72**	-.20**	—							
4. Neuroticism	2.74	0.94	-.16**	.04**	-.20**	(.82)						
5. Anger	0.74	0.68	-.09**	.06**	-.13**	.43**	(.94)					
6. Career calling	2.16	0.52	.10**	-.13**	.02**	-.12**	-.04**	(.89)				
7. POQ	3.43	0.59	-.50**	.14**	-.13**	.15**	.12**	-.06**	(.86)			
8. Job boredom	2.94	1.08	-.18**	.16**	-.15**	.34**	.30**	-.15**	.33**	(.85)		
9. CWB	2.45	0.98	.11**	-.17**	.08**	-.27**	.31**	.11**	.12**	.28**	(.93)	
10. OCB	4.24	0.93	.10**	-.26**	.15**	-.17**	-.17**	.41**	-.09**	-.27**	.06*	(.92)

Note. *N* = 298. Gender (0 = male, 1 = female) is a dummy variable. A unit of job tenure is a month. POQ = perceived overqualification; CWB = counterproductive work behavior; OCB = organizational citizenship behavior.

p* < .05. *p* < .01.

Table 2. Summary Data for Moderated Mediation Models.

Models	Output
Model 1: Job boredom → CWB	$R^2 = .23, F(297) = 10.60, p < .001$
Job boredom	.21*** (.05)
Career calling	.26* (.10)
Job Boredom × Career Calling	.07 (.09)
Model 2: Job boredom → OCB	$R^2 = .29, F(297) = 14.38, p < .001$
Job boredom	-.14** (.05)
Career calling	.65*** (.09)
Job Boredom × Career Calling	.17* (.08)
Model 3: POQ → CWB	$R^2 = .23, F(9, 288) = 9.47, p < .001$
POQ	.07 (.09)
Job boredom	.20*** (.05)
Index of moderated mediation (indirect effect)	.03 (.04), CI [-.0552, .1233]
Model 4: POQ → OCB	$R^2 = .53, F(9, 288) = 12.81, p < .001$
POQ	.06 (.09)
Job boredom	-.15** (.05)
Index of moderated mediation (indirect effect)	.08 (.05), CI [.0008, .1910]

Note. *N* = 298. Unstandardized coefficients are reported, and standard errors are reported in the parentheses.

POQ = perceived overqualification; CWB = counterproductive work behavior; OCB = organizational citizenship behavior.

p* < .05. *p* < .01. ****p* < .001.

between POQ and CWB (indirect effect = .08, 95% CI [.04, .20]) and OCB (indirect effect = -.06, 95% CI [-.19, -.02]), supporting Hypotheses 4a and 4b.

As a part of our moderated mediation model, we investigated whether career calling moderates the effect of job boredom on CWB and OCB. As shown in Table 2, career calling did not significantly moderate the relation between job boredom and CWB ($\beta = .04, p = .46$). Therefore, Hypothesis 5a was not supported, and this made Hypothesis 6a unsupported as well (Hayes, 2015). Hypothesis 5b was supported as career calling significantly moderated the relation between job boredom and OCB ($\beta = .09, p < .05$), such that among those whose career calling was relatively high (vs. low), the negative relation between job boredom and OCB became weaker (vs. stronger). Simple slope tests further

revealed that when career calling was low (-1 *SD* from the mean), job boredom significantly predicted OCB ($\beta = -.26, p < .001$). Conversely, when career calling was high ($+1$ *SD* from the mean), job boredom was not significantly related to OCB ($\beta = -.06, p = .40$).

Finally, a formal test of moderated mediation showed that the indirect effect of POQ on OCB through job boredom was significant for employees with low career calling (conditional indirect effect = $-.12$, 95% bias-corrected CI [$-.22, -.04$]). In contrast, among those with high career calling, this indirect effect was not significant (conditional indirect effect = $-.03$, 95% bias-corrected CI [$-.11, .03$]). The difference among the conditional indirect effects was significant as a bootstrap CI for the index of moderated mediation excluded zero (index of moderated mediation = $.08$, 95% CI [$.0008, .19$]). Thus, Hypothesis 6b was supported, indicating that the indirect effect of POQ via job boredom was contingent on level of career calling.

Discussion

Our research makes several important contributions to the literature. First, viewing POQ as P-J misfit, we examined the mediating role of job boredom in the relations between POQ and CWB and OCB. Notably, even after controlling for state anger, neuroticism, and demographic variables (age, gender, and job tenure), we found that job boredom served as an affective mechanism in the underlying process. This finding extended studies of Luksyte, Spitzmueller, and Maynard (2011) and Liu, Luksyte, Zhou, Shi, and Wang (2015) by replicating the overall positive relation between POQ and CWB among South Korean employees. Our findings also contribute to P-J fit theory in that overqualified individuals who feel the misfit are indeed less likely to engage in more CWB and perform less OCB since they feel bored at work.

The study results are in line with the emotion-centered model of CWB and OCB (Spector & Fox, 2002) in that they demonstrated that South Korean employees who experience the negative emotion (i.e., job boredom) in the presence of stressor (i.e., POQ) engage in behaviors detrimental to the organization (i.e., CWB). Our findings support the theoretical propositions of Liu and Wang (2012), who argue that boredom can play a key role in mediating the associations between overqualification and maladaptive work behaviors (i.e., CWB). The results are also consistent with those of prior work; overqualified workers tend to engage in CWB (Liu et al., 2015; Luksyte et al., 2011) and are less likely to demonstrate interpersonal altruism and team member proactivity (Deng et al., 2018) due to increased negative emotions (Belschak & Den Hartog, 2009; Liu et al., 2015). With regard to OCB of overqualified workers, Deng et al. (2018) have shown that, when experiencing negative affect as a consequence of POQ, these employees interacted with coworkers or supervisors in a rude manner that damaged their social relationships. Our findings suggest that when overqualified workers feel bored, their level of extra-role performance may be poorer than less qualified employees. Our results expand the literature on POQ and extra-role behaviors by showing that low-arousal negative emotions such as boredom can play a critical role in understanding the CWB and OCB of overqualified workers.

Our study shows that boredom is negatively related to OCB. Although some scholars suggest that bored employees may help others or engage in other activities that can be helpful to the organization in order to avoid boredom (Fisher, 1993; Spector & Fox, 2010), our findings are in line with the argument that negative emotions such as boredom decrease OCB. Given that the concept of job boredom is not limited to work monotony or task repetitiveness but also encompasses one's affective response to the work environment (Game, 2007), bored employees may find both tasks and people tedious and understimulating, which could be a possible explanation for the negative link between job boredom and OCB. Recent work by Brodsky and Amabile (2018) also supports our findings. Using a series of six studies, they investigated the consequences of involuntary idle time at work and found that workers who expected involuntary idle time intentionally slowed work pacing. This supports the notion

that employees may respond to boredom in an undesirable manner rather than engaging in altruistic behaviors.

Our finding related to career calling offers some implications for the literature of career calling and organizational psychology. Having a career calling generally benefits both the worker and the organization (Duffy & Dik, 2013) because people with callings focus on noble ends of their work with prosocial intentions (Elangovan et al., 2010). We demonstrated that career calling weakened the negative relation between job boredom and OCB. This finding suggests that despite boring work and job conditions, employees with high career calling may still be motivated to perform better by demonstrating OCB. The findings are consistent with prior findings that show employees whose work is seen as a calling are known to excel at OCB (Park et al., 2016) and that they show greater commitment to the organization in pursuit of greater good and meaningfulness through their work (Bunderson & Thompson, 2009).

Contrary to our expectations, career calling did not moderate the relation between job boredom and CWB. The findings showed that bored workers engage in counterproductive behaviors, regardless of their level of career calling. This finding may be partly due to the fact that workers with career calling may react to negative situations more intensely because of high identification with their work (Schabram & Maitlis, 2017). People with career calling become deeply involved with their work and set high expectations for their job performance (Schabram & Maitlis, 2017), and when reality contradicts their expectations, it may result in negative emotions and deviant behaviors. Also, the nonsignificant relation between career calling and CWB suggests that career calling, a positive psychological construct (Duffy & Dik, 2013), may be more relevant to promoting positive work behaviors such as OCB and less relevant to preventing negative organizational behaviors such as CWB.

From a practical point of view, the results can provide insights for career counselors and supervisors. Our finding that boredom relates to adverse outcomes at work indicates that it is important to design jobs in a way to prevent boredom. Supervisor and executives should assign less monotonous tasks but more challenging ones to overqualified employees. Career counselors also should recommend their clients that they perform job crafting behaviors to decrease boredom. Job crafting behaviors such as increasing structural resources were found to be effective in reducing the effects of job boredom on CWB (van Hooff & van Hooft, 2014). Second, findings that calling moderated the indirect effect of POQ on OCB through job boredom suggest that cultivating and perceiving career calling can be beneficial to people with POQ and organizations. Counselors can help their clients identify and develop career calling by facilitating their growth of self-awareness (Elangovan et al., 2010) or by providing calling workshops (e.g., Dik & Steger, 2008).

Limitations and Suggestions for Future Research

Our study is subject to some limitations that suggest directions for future research. First, we used self-report measures of the study variables in a single survey. Using a cross-sectional design can be restrictive when inferring causal relations of the variables. It is also possible that career calling weakens the links from POQ to CWB and OCB, suggesting an interaction between POQ and career calling on the two behaviors. Future research could employ a longitudinal design and an experience-sampling method to gain a better understanding of the relations among the study variables.

Second, our findings are based on young white-collar employees in East Asian cultures. Future research is warranted to increase the generalizability of our findings to older groups, other occupations, and other sociocultural contexts. Experience of boredom varies depending on demographic variables such as gender and age (Chin, Markey, Bhargava, Kassam, & Loewenstein, 2017), and a high portion of young employees perceive themselves as overqualified (Harari et al., 2017). The emotional experience is influenced by sociocultural contexts, and propensity for boredom varies as a function of social surroundings and contexts (Chin et al., 2017). In this study, we did not include

employees at advanced career stages or examine how sociocultural or organizational contexts impact relations between study variables.

Lastly, further attention to the characteristics of boredom that account for the effects of boredom on CWB and OCB is warranted. Boredom co-occurs with other negative emotions, such as loneliness and anger (Chin et al., 2017). This study suggests that unique characteristics of boredom that distinguish from anger influence CWB and OCB. Although boredom primarily arises from task characteristics such as monotonous or repetitive tasks, individuals also experience boredom in interpersonal relations (Leary, Rogers, Canfield, & Coe, 1986). Given that the adverse effects of boredom extend to interpersonal behaviors such as OCB and CWB, the experience of boredom can be influenced by both task and interpersonal domains. Thus, how boredom interacts with different domains in exerting its negative effects warrants future research.

Conclusion

This study contributes to the existing literature by identifying a critical role of job boredom that explains paths from POQ to OCB and CWB. Our study showed that job boredom fully mediates the relations of POQ with CWB and OCB, where career calling moderates the relation between job boredom and OCB. The results have implications for practitioners that creating an environment that reduces boredom and fulfills career calling can lead to more OCB and less CWB. Although the present study has limitations that highlight future research, the findings of this study provide a contribution that might guide improved career counseling and continued research.

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